

Enhancing Foundational Numeracy in Tribal Students via Technology and Community Facilitators: A Quasi-Experimental Study

Amod Joshi¹, Saarang Rajguru¹, Mrinmoyi Kulkarni², and Milind Kelkar³

¹ Nplusone Social Foundation, Pune, India,
`amod.m.joshi@gmail.com`,

² Department of Humanities and Social Sciences, Indian Institute of Technology
Bombay, Mumbai, India,
`mrinmoyi@iitb.ac.in`,

³ Finearc Systems, Pune, India
`mkelkar@finearcsystems.com`

Abstract. Foundational numeracy is critical for academic and vocational success across all disciplines, yet students in India’s tribal regions often lack these essential skills, limiting their educational outcomes. This paper presents the results of a quasi-experimental study evaluating a hybrid or “blended” intervention, named “Solve-with-Bharat”, combining a mobile-based math application with community facilitator support through in-school practice sessions for foundational numeracy skills among 1,000 students (Grades 3-6) in Zilla Parishad schools in Talasari taluka, Palghar district, Maharashtra.

Over six months, the study tracked 500 treatment group students and measured foundational numeracy improvements, showing substantial gains with an effect size of 1.17 ($p < 0.01$). These findings suggest that hybrid approaches offer scalable solutions to India’s foundational learning crisis, particularly in underserved tribal communities where human engagement amplifies technological interventions.

Keywords: Foundational Numeracy, Educational Technology, Rural Education, Low-cost Interventions

1 Introduction

1.1 Problem Context and Motivation

India’s NEP 2020 aims for universal foundational numeracy (FN) by 2025, yet many students—especially in tribal Zilla Parishad schools—struggle with basic operations such as addition, subtraction, multiplication, and division. Only 36% of Grade 8 students in rural Maharashtra can perform division [1]. Contributing factors include language mismatch [2], poor teacher-pupil ratios, and teacher duties outside academics [3].

1.2 Existing Work and Need For Solution

Past research shows that Teaching at the Right Level (TaRL) [4] and Computer-Aided Learning (CAL) [5] improve learning, but most rural programs face resistance or infrastructure challenges [6, 4].

Interventions like the one implemented by Indian education firm Educational Initiatives studied by Muralidharan et al. [5] show us that hybrid or “blended” learning approaches combining Computer-Aided Learning (CAL) with human facilitation can produce significant gains. However, CAL interventions required resource-rich centers and urban settings, limiting rural applicability.

The Solve-with-Bharat model combines these approaches - mobile-based, level-appropriate math practice guided by trained local community facilitators (CFs) and affordable mobile-based CAL technology - to deliver regular, retrieval-based practice within school hours.

1.3 Approach

The Solve-with-Bharat program leverages a combination of technology and human support. The core idea is that practice leads to perfection, and foundational math is mastered through regular doing and retrieval-based learning. The solution involves community facilitators (CFs) that are local graduates or college students, mobile phones and practice sessions on them delivered by the CFs.

1.4 Research Objective and Questions

1. Does the Solve-with-Bharat intervention significantly improve foundational numeracy in Grades 3–6 tribal students?
2. What is the effect size of this improvement?
3. Are gains consistent across arithmetic operations?
4. Is the model cost-effective and scalable?

1.5 Summary of Key Results

The experiment demonstrated a statistically significant improvement in foundational numeracy skills for the treatment group compared to the control group, achieving a large effect size ($d = 1.17$ over six months). Further details are presented in Section 5 (Results).

1.6 Contributions

This research provides robust evidence for a cost-effective “blended” intervention model combining technology and local community facilitators to achieve significant learning gains in foundational numeracy among students in tribal areas. Beyond student outcomes, the program notably contributes to the development and employability of local community members serving as facilitators, enhancing their digital skills and providing a livelihood.

2 Related Work

2.1 Theoretical Basis

The intervention’s design is based on cognitive learning theories that emphasize the importance of retrieval practice and repetition for long-term retention and fluency. Roediger and Butler [7] have shown that retrieval practice produces large gains in long-term retention relative to repeated studying, especially when combined with timely feedback.

Another important theoretical aspect of the intervention is the TaRL approach, which is supported by numerous impact evaluations, randomized controlled trials (RCTs), and global studies [4].

2.2 Related Work

Pratham’s Balsakhi program and other TaRL experiments achieved measurable improvements by grouping students by ability rather than grade [8]. However, implementation within government schools often faces institutional barriers [9, 6].

Urban CAL projects such as Mindspark and OneBillion [10, 5] produced gains but required expensive infrastructure, limiting scalability. After-school TaRL programs improved outcomes but suffered from low attendance. [6]

Therefore, existing solutions are either technology-intensive and difficult to scale to rural areas, or community-based but without technological support for systematic competency building and progress tracking. The Solve-with-Bharat program proposes a solution that directly addresses the identified gaps by leveraging CFs with accessible and affordable mobile-based CAL technology during school hours, increasing both personalized attention and practice.

3 Implementation

3.1 The Intervention: Community Facilitator and Mobile App Model

The program operated in 15 Zilla Parishad schools in Talasari taluka, Palghar district, covering Grades 3–6. 500 students from 8 randomly selected schools formed the treatment group and 500 students from comparable schools nearby formed the control.

Community Facilitators (CFs): Local graduates received short training in technology and pedagogy. Each CF managed 20–40 students per session, holding three sessions per week using mobile phones loaded with the math practice app. Students solved around 420 sums per week, sharing devices in small groups.

CFs logged progress after every session. Across all students, the program achieved 75.8% efficiency (617,544 sums completed weekly against a 814,380 target). Five assessments—baseline, three formative, and one endline—measured progress.

3.2 Cost Effectiveness

The total cost of the program over a 9-month period amounted to INR 20,68,000, out of which mobile phone depreciation costs (assuming 30% depreciation) were INR 7,00,000 and staff salaries amounted to INR 13,68,000. The per-child cost for the program was INR 1,034 over 9 months, which averages to INR 115 per month per student.

4 Research Method

4.1 Research Design and Sample

To establish causality, a two-group pretest-posttest quasi-experiment was conducted with 1000 students (Grades 3-6): 500 in the treatment group from 8 randomly selected newly enrolled Z.P. schools and 500 controls from 3 Z.P. schools in the same area, ensuring comparable socioeconomic status (SES). Random selection of new schools minimized familiarity and confounding factors, while requiring full school participation avoided exclusion effects. Power calculations for a t-test of significance ($\alpha = 0.05$, Cohen's $d = 0.2$, power = 0.8) indicated 393 students per group. This figure was padded to 500 to ensure robustness. Baseline evaluation confirmed equivalent starting levels.

4.2 Procedure

Students were benchmarked monthly over a 6-month period, using an identical 32-sum test offered in English and Marathi, covering eight specific categories of foundational math operations to ensure consistent measurement of progress. The categories included: addition with no carry-over, addition with carry-over, subtraction with no borrow, subtraction with borrow, simple multiplication facts, harder multiplication facts, multiplication and division. One hour was allotted for completion. CFs proctored and graded the assessments. The use of the same assessment for all evaluations across baseline, formative, and endline periods ensured test-retest reliability.

4.3 Data Analysis Techniques

A Welch Two-Sample t-test ($\alpha = 0.01$) was conducted using the R statistical program to compare the means of the treatment and control groups, and effect size was quantified using Cohen's d .

4.4 Ethical Considerations

Formal permission was taken for the program from the Principal of each school as well as from the respective Cluster Head and Block Education Officer (BEO). Parental consent was meticulously collected from parents of all students participating in the program. The quasi-experimental approach ensured fairness and avoided exclusion effects while maintaining experimental validity.

5 Results

5.1 Quasi-Experiment Results

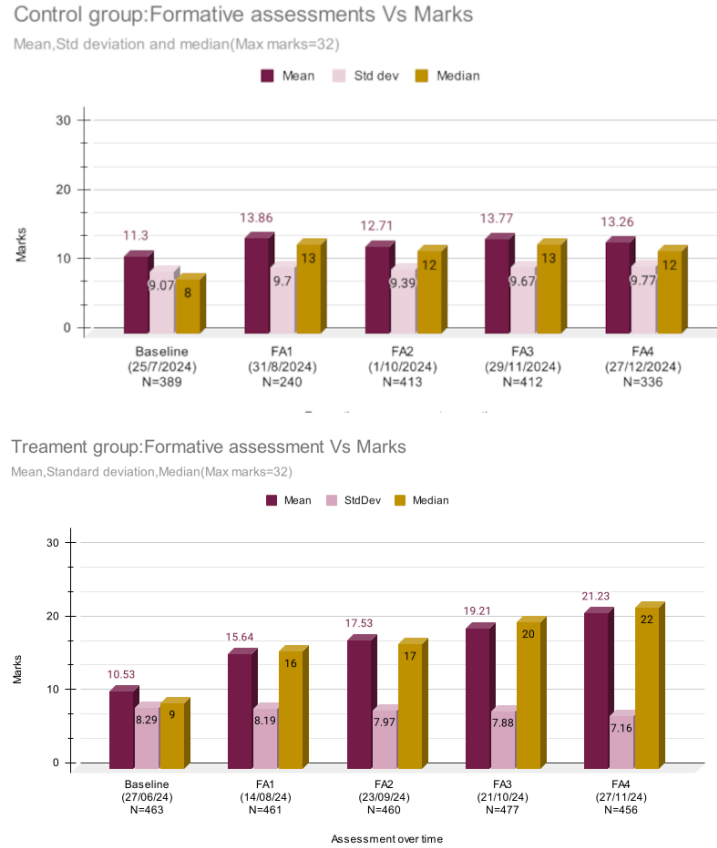


Fig. 1. Evaluation performance of the treatment group and control group.

The Solve-with-Bharat intervention demonstrated a significant and substantial learning gain in the treatment group over the control group. Over a 6-month period, the mean score in the treatment group improved 2.13 times from baseline, and the median score improved 2.6 times (as seen in Fig. 1). In contrast, the control group showed only modest improvement over the same 6-month period, with its mean improving just 1.2 times and its median 1.5 times from baseline (as seen in Fig. 2). Notably, the standard deviation in the treatment group reduced significantly from 8.29 at baseline to 5.24 by the end of the intervention, indicating increased consistency and less variation in student performance.

5.2 Statistical Analysis

The Welch Two Sample t-test resulted in a t-statistic of 24.399 and a p-value $p < 0.01$. Given a 1% significance level, the null hypothesis ($\mu_1 = \mu_2$) was rejected in favor of the alternative hypothesis ($\mu_1 > \mu_2$), providing strong evidence that the treatment mean score is greater than the control mean score. Cohen's d effect size was estimated at 1.17, interpreted as a large effect size. The 99% confidence interval for the mean difference was $[12.52055, \text{Inf}]$, implying high certainty that the treatment mean is at least 12.52 marks greater than the control mean.

5.3 Competency and Student Improvement

Students improved across all eight categories of foundational operations. The mean and median increased while the spread decreased, indicating improved performance across the group, as seen in Fig. 3a. Crucially, the percentage of students who improved in the treatment group reached close to 100% by the end of the evaluation period.

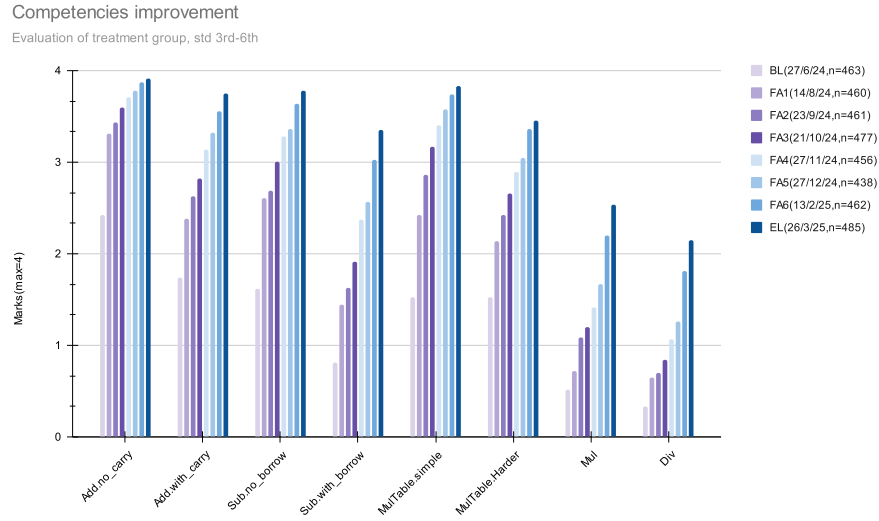


Fig. 2. Percentage of students showing improvement during the program.

6 Discussion

6.1 Answering Research Questions

Based on statistically significant results, large effect size and consistent improvement in multiple metrics, the study concludes that the Solve-with-Bharat inter-

vention caused a significant improvement in foundational numeracy skills in the target population, answering the primary research question regarding causality. The magnitude of this effect is large ($d=1.17$). The program led to improvements in specific operations, indicating a broad impact on foundational understanding.

6.2 Connection to Prior Work

The findings from the Solve-With-Bharat program support the core assumption that regular practice and personalized, retrieval-based learning leads to improved learning outcomes, particularly for foundational skills. This also aligns with and provides valuable confirmation for the TaRL approach, which emphasizes instruction tailored to a student’s actual learning level (see “Related Work”).

This intervention also addresses the challenges faced by technology-intensive interventions, such as those by Mindspark and OneBillion, which, despite showing significant gains, often faced sustainability issues due to cost or low turnout in after-school centres. By embedding accessible technology (mobile phones) within existing school structures and leveraging local CFs, Solve-with-Bharat overcomes logistical hurdles and provides a sustainable, scalable model for rural contexts where traditional computer infrastructure is not feasible.

6.3 Claims and Generalizability

The generalizability of the findings is argued based on the random selection of schools within the target area. The success across multiple schools and with different CFs suggests that the effect is not school-specific or dependent on particular individuals, but rather an outcome of the program’s systematic design and implementation. Hence, the model is believed to be generalizable to similar tribal and rural settings facing comparable educational challenges across India and potentially other developing countries.

6.4 Limitations and Validity Threats

The study faced the following limitations and challenges:

1. **School Population Constraints:** While ZP schools widely vary in size (some schools having less than 20 students, while others have 200+), the program targeted schools that have a large population (around 100) to ensure that CFs could reach a sufficient number of schools, as most CFs have limited commute options.
2. **Absence of triangulation with qualitative data:** The program focused on quantitative metrics to determine learning gains. Since the intervention involved multiple components (mobile app, CF facilitation, regular practice), a limitation is the lack of qualitative data (such as in-depth interviews with students, teachers or CFs) to triangulate and provide a richer understanding of the observed quantitative results.

7 Conclusion

The Solve-with-Bharat program demonstrates a powerful and cost-effective model to improve foundational numeracy in resource-constrained tribal schools by leveraging mobile technology and local community facilitators. The quasi experimental design provided strong evidence for the causal impact and magnitude of this intervention, resulting in significant learning gains in a short period.

This work contributes to a validated “blended” intervention model to address critical foundational skill gaps in challenging educational environments, particularly for first-generation learners in rural and tribal regions. Future research should focus on scaling the program to reach more students and refining CF training based on the lessons learned. Further studies could explore the long-term impact on student achievement and the generalizability of the model to diverse geographies and educational levels. The findings suggest that focused practice interventions, supported by appropriate technology and human guidance, have significant potential to level the educational playing field.

References

1. ASER Centre: Aser 2024 survey (maharashtra - rural) (2024)
2. Bühmann, D.: Mother tongue matters: Local language as a key to effective learning. UNESCO (2008)
3. World Bank: World Development Report 2018: Learning to Realize Education’s Promise. Washington, DC: World Bank (2017). doi:10.1596/978-1-4648-1096-1
4. Banerjee, A., Banerji, R., Berry, J., Duflo, E., Kannan, H., Mukherji, S., Shotland, M., Walton, M.: Mainstreaming an effective intervention: Evidence from randomized evaluations of “teaching at the right level” in india. Tech. rep., National Bureau of Economic Research (2016). doi:10.3386/w22746
5. Muralidharan, K., Singh, A., Ganimian, A.J.: Disrupting education? experimental evidence on technology-aided instruction in india. *American Economic Review* 109(4), 1426–1460 (2019). doi:10.1257/aer.20171112
6. Banerjee, A., Banerji, R., Berry, J., Duflo, E., Kannan, H., Mukerji, S., Shotland, M., Walton, M.: From proof of concept to scalable policies: Challenges and solutions, with an application. *Journal of Economic Perspectives* 31(4), 73–102 (2017). doi:10.1257/jep.31.4.73
7. Roediger, H.L., Butler, A.C.: The critical role of retrieval practice in long-term retention. *Trends in cognitive sciences* 15(1), 20–27 (2011). doi:10.1016/j.tics.2010.09.003
8. Duflo, E., Dupas, P., Kremer, M.: Peer effects, teacher incentives, and the impact of tracking: Evidence from a randomized evaluation in kenya. *American Economic Review* 101(5), 1739–1774 (2011). doi:10.1257/aer.101.5.1739
9. Betts, J.R.: The economics of tracking in education. In: *Handbook of the Economics of Education*, vol. 3, pp. 341–381. Elsevier (2011). doi:10.1016/B978-0-444-53429-3.00007-7
10. Levesque, K., Bardack, S., Chigeda, A.: Tablet-based learning for foundational literacy and math: an 8-month rct in malawi. *Res. Rep. Prep. Imag. Worldw* (2020)